

# Automotive Wholesale & Resale Price Dynamics

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## Context and Questions

### Background & Context

The automobile industry represents a critical segment of the global economy, linking automotive manufacturers (OEMs), fleet leasing corporations, wholesale auto auctions, franchised/independent dealerships, and retail consumers. Amid shifting macroeconomic conditions—including political misalignment, supply chain issues, and stiff competition—used vehicle values serve as a leading indicator for automotive brand equity and implications for stakeholders. The primary actors within this domain include OEM sellers (e.g., captive finance arms such as Ford Motor Credit or Kia Motors America remarketing off-lease vehicles), auction remarketers, and used vehicle dealerships acquiring inventory. In wholesale auction environments, transactions are benchmarked against industry wholesale standards, notably the Manheim Market Report (MMR) value, which estimates wholesale value based on recent actual wholesale sales. This project investigates transactional wholesale vehicle sales to uncover the key vehicular, temporal, and market factors that determine realized transaction prices.

### Business Problem & Opportunity

Managing pre-owned inventory and predicting asset depreciation pose significant operational and financial risks for automotive remarketers, lenders, and dealerships. For leasing and financing companies, inaccurate residual value forecasting directly impacts lease portfolio write-downs. For automotive dealerships, mispricing wholesale purchases can lead to margin compression, stagnant inventory, and holding costs. Investigating what attributes drive final selling prices—and identifying where transaction prices systematically deviate from baseline benchmarks like MMR—presents an opportunity to optimize inventory pricing, procurement timing, and remarketing strategies.

### Initial Business Questions

1. **Primary Determinants of Transaction Value:** How do physical vehicle attributes (model year, mileage/odometer reading, mechanical/cosmetic condition score, body style, and make) interact to explain variance in final selling price?
2. **Benchmark Deviation (MMR vs. Actual Sale Price):** Under what conditions (e.g., specific vehicle age, geographic market/state, or seller type) do final selling prices systematically deviate from the Manheim Market Report (MMR) baseline estimate?
3. **Actionable Value:** Answering these questions enables dealership managers and remarketers to identify undervaluation patterns, purchase wholesale inventory more profitably, and optimize the liquidation timing of off-lease fleet assets.

# Section 2: Data and Variables

## Source and Acquisition

The dataset was obtained from Kaggle under the title *Vehicle Sales Data*, compiled by Syed Anwar Afridi from publicly shared automotive auction transaction data. The data reflects real-world US auto auction transactions (predominantly Manheim auction feeds) spanning late 2014 through mid-2015. These data directly address the proposed business questions because they record actual finalized sale transactions rather than advertised listing prices, paired directly with industry valuation benchmarks (MMR) and standardized multi-point physical vehicle condition assessments.

## Data Grain and Dataset Dimensions

For the data grain, one row in the dataset represents a single wholesale auction transaction of a specific used vehicle, uniquely identified by its 17-character Vehicle Identification Number (VIN). The dataset contains 558,837 observations (rows) and 16 variables (columns), comfortably exceeding academic guidelines for either statistical modeling or BI dashboard development.

## Main Variables Description

| Vehicle Sales Data |          |                     |                           |             |              |                  |       |           |          |       |          |                                                     |       |              |                                         |
|--------------------|----------|---------------------|---------------------------|-------------|--------------|------------------|-------|-----------|----------|-------|----------|-----------------------------------------------------|-------|--------------|-----------------------------------------|
| year               | make     | model               | trim                      | body        | transmission | vin              | state | condition | odometer | color | interior | seller                                              | mmr   | sellingprice | saledate                                |
| 2015               | Kia      | Sorento             | LX                        | SUV         | automatic    | 5ytcnablg64472   | ca    | 5         | 18839    | white | black    | ku motors america inc                               | 20500 | 21500        | Tue Dec 16 2014 12:30:00 GMT-0800 (PST) |
| 2015               | Kia      | Sorento             | LX                        | SUV         | automatic    | 5ytcnablg645138  | ca    | 5         | 9593     | white | beige    | ku motors america inc                               | 20800 | 21500        | Tue Dec 16 2014 12:30:00 GMT-0800 (PST) |
| 2014               | BMW      | 3 Series            | 328i xDrive               | Sedan       | automatic    | whb311514613051  | ca    | 45        | 1331     | gray  | black    | francois services remarketing (hawaii)              | 31900 | 30500        | Thu Jan 16 2015 04:30:00 GMT-0800 (PST) |
| 2015               | Vaux     | SRZ                 | 16                        | Sedan       | automatic    | yx43234471520881 | ca    | 42        | 24282    | black | black    | valley va repurchase center                         | 27500 | 27750        | Thu Jan 15 2015 04:30:00 GMT-0800 (PST) |
| 2014               | BMW      | 4 Series Gran Coupe | 450i                      | Sedan       | automatic    | whb425746128721  | ca    | 43        | 28451    | gray  | black    | francois services remarketing (hawaii)              | 68000 | 67000        | Thu Dec 18 2014 12:30:00 GMT-0800 (PST) |
| 2015               | Nissan   | Altima              | 2.5 S                     | Sedan       | automatic    | 2k43dab161520212 | ca    | 1         | 5554     | gray  | black    | enterprise vehicle exchange / tra / rental / fuelco | 15500 | 15000        | Tue Dec 16 2014 12:30:00 GMT-0800 (PST) |
| 2014               | BMW      | HS                  | Base                      | Sedan       | automatic    | whb425746153080  | ca    | 34        | 14853    | black | black    | the hertz corporation                               | 69000 | 65000        | Wed Dec 17 2014 12:30:00 GMT-0800 (PST) |
| 2014               | Chrysler | Charger             | 157                       | Sedan       | automatic    | 2g1fb16426153460 | ca    | 2         | 28817    | black | black    | enterprise vehicle exchange / tra / rental / fuelco | 15800 | 16000        | Tue Dec 16 2014 12:30:00 GMT-0800 (PST) |
| 2014               | Audi     | A4                  | 2.0T Premium Plus quattro | Sedan       | automatic    | wu1ff346330343   | ca    | 42        | 9557     | white | black    | audi mission msp                                    | 32100 | 32250        | Thu Dec 18 2014 12:30:00 GMT-0800 (PST) |
| 2014               | Chrysler | Camero              | 157                       | Convertible | automatic    | 2g1fb16426153460 | ca    | 3         | 4809     | red   | black    | dmv auto sales inc                                  | 26300 | 17000        | Tue Jan 13 2015 04:30:00 GMT-0800 (PST) |
| 2014               | Audi     | A4                  | 2.0T Premium Plus quattro | Sedan       | automatic    | wu1ff346330343   | ca    | 48        | 14414    | black | black    | desert auto trade                                   | 47300 | 49750        | Thu Dec 18 2014 12:30:00 GMT-0800 (PST) |
| 2014               | Kia      | Optima              | LX                        | Sedan       | automatic    | 5ytcnablg641467  | ca    | 40        | 2014     | red   | tan      | ku motors america inc                               | 15150 | 17750        | Tue Dec 16 2014 12:30:00 GMT-0800 (PST) |
| 2014               | Ford     | Fusion              | SE                        | Sedan       | automatic    | 3f4p4d4671467513 | ca    | 2         | 5559     | white | beige    | enterprise vehicle exchange / tra / rental / fuelco | 15350 | 12000        | Tue Jan 13 2015 12:30:00 GMT-0800 (PST) |
| 2015               | Kia      | Sorento             | LX                        | SUV         | automatic    | 5ytcnablg641467  | ca    | 5         | 14814    | black | black    | ku motors america inc                               | 20800 | 21500        | Tue Dec 16 2014 12:30:00 GMT-0800 (PST) |
| 2014               | Chrysler | Charger             | 237                       | Sedan       | automatic    | 3g1p4d467120897  | ca    | 3         | 10886    | black | black    | ku motors america inc                               | 15800 | 15000        | Thu Dec 18 2014 12:30:00 GMT-0800 (PST) |
| 2015               | Nissan   | Altima              | 2.5 S                     | Sedan       | automatic    | 2k43dab161520212 | ca    | 2         | 11186    | black | black    | and fcp                                             | 14750 | 14100        | Tue Dec 16 2014 12:30:00 GMT-0800 (PST) |
| 2014               | Hyundai  | Elantra             | SE                        | Sedan       | automatic    | 5ytcnablg641467  | ca    | 8511      | red      | black | black    | enterprise vehicle exchange / tra / rental / fuelco | 12000 | 4100         | Tue Dec 16 2014 12:30:00 GMT-0800 (PST) |
| 2014               | Audi     | Q5                  | 2.0T Premium Plus quattro | SUV         | automatic    | wu1ff346330343   | ca    | 49        | 7963     | white | black    | audi north carolina                                 | 37100 | 40000        | Thu Dec 18 2014 12:30:00 GMT-0800 (PST) |
| 2014               | Chrysler | Camero              | 157                       | Convertible | automatic    | 2g1fb16426153460 | ca    | 17        | 13461    | black | black    | webb kapp motor services                            | 17750 | 17000        | Tue Dec 16 2014 12:30:00 GMT-0800 (PST) |
| 2014               | BMW      | 4 Series            | 450i                      | Convertible | automatic    | whb425746153080  | ca    | 34        | 8819     | black | black    | the hertz corporation                               | 68000 | 67200        | Wed Dec 17 2014 12:30:00 GMT-0800 (PST) |
| 2015               | Chrysler | Impala              | 157                       | Sedan       | automatic    | 2g1fb16426153460 | ca    | 39        | 14538    | black | black    | enterprise vehicle exchange / tra / rental / fuelco | 14800 | 7500         | Tue Jan 13 2015 09:30:00 GMT-0800 (PST) |
| 2014               | BMW      | 4 Series            | 450i                      | Sedan       | automatic    | whb425746153080  | ca    | 29        | 25909    | black | black    | francois services remarketing (hawaii)              | 34200 | 30500        | Tue Feb 10 2015 04:30:00 GMT-0800 (PST) |
| 2014               | Chrysler | Camero              | 157                       | Convertible | automatic    | 2g1fb16426153460 | ca    | 34        | 23400    | black | black    | ku motors america inc                               | 20100 | 14750        | Tue Dec 16 2014 12:30:00 GMT-0800 (PST) |
| 2015               | Audi     | A3                  | 1.8 TFSI Premium          | Sedan       | automatic    | wu1ff346330343   | ca    | 49        | 5836     | gray  | black    | audi north carolina                                 | 24000 | 23750        | Thu Dec 18 2014 12:30:00 GMT-0800 (PST) |
| 2014               | BMW      | 4 Series            | 450i                      | Convertible | automatic    | whb425746153080  | ca    | 38        | 12736    | black | black    | the hertz corporation                               | 67000 | 68000        | Tue Dec 16 2014 12:30:00 GMT-0800 (PST) |
| 2015               | Hyundai  | Sonata              | SE                        | Sedan       | automatic    | 5ytcnablg641467  | ca    | 9291      | white    | gray  | gray     | enterprise vehicle exchange / tra / rental / fuelco | 15150 | 8500         | Thu Dec 18 2014 12:30:00 GMT-0800 (PST) |
| 2015               | Vaux     | ACZT                | 16                        | Wagon       | automatic    | yx43234471520881 | ca    | 42        | 10500    | brown | black    | valley va repurchase center                         | 21200 | 22500        | Tue Feb 10 2015 04:30:00 GMT-0800 (PST) |
| 2015               | Vaux     | ACZT                | 16                        | Wagon       | automatic    | yx43234471520881 | ca    | 40        | 12775    | gray  | beige    | valley va repurchase center                         | 22200 | 25000        | Thu Feb 12 2015 04:30:00 GMT-0800 (PST) |
| 2014               | BMW      | 4 Series            | 450i                      | Sedan       | automatic    | whb425746153080  | ca    | 42        | 11278    | gray  | black    | ku motors america inc                               | 15400 | 16000        | Tue Dec 16 2014 12:30:00 GMT-0800 (PST) |
| 2014               | Chrysler | Charger             | 157                       | Convertible | automatic    | 2g1fb16426153460 | ca    | 42        | 11474    | gray  | black    | midway inc/peppers                                  | 22200 | 15000        | Thu Dec 18 2014 12:30:00 GMT-0800 (PST) |
| 2014               | Chrysler | Charger             | 157                       | Convertible | automatic    | 2g1fb16426153460 | ca    | 42        | 28818    | gray  | black    | enterprise vehicle exchange/longrange               | 15800 | 15500        | Tue Dec 16 2014 12:30:00 GMT-0800 (PST) |
| 2014               | Audi     | A6                  | 2.0T Premium Plus quattro | SUV         | automatic    | wu1ff346330343   | ca    | 42        | 9606     | white | black    | audi of downtown la                                 | 48800 | 47000        | Thu Jan 15 2015 04:30:00 GMT-0800 (PST) |
| 2015               | Kia      | Sorento             | LX                        | SUV         | automatic    | 5ytcnablg645138  | ca    | 5         | 12757    | red   | black    | ku motors america inc                               | 20500 | 20750        | Tue Dec 16 2014 12:30:00 GMT-0800 (PST) |

The dataset contains sixteen variables organized into the four broad categories outlined in the project guidelines: identifiers, categorical variables, dates and time fields, and numeric measures. The primary identifier in the data is vin, which represents the unique 17-character Vehicle Identification Number assigned to each transaction. The categorical variables capture descriptive vehicle attributes, market context, and cosmetic specifications. Vehicle specifications include make (manufacturer brand), model (vehicle product line), trim (sub-model or trim package), body (vehicle body configuration), and transmission (gearbox type). Styling choices are captured by color and interior, reflecting exterior paint and cabin upholstery colors, respectively. In addition, transaction logistics are recorded through state, which indicates the two-letter state code of the auction location, and seller, which designates the liquidating organization or dealership. The dates and time fields provide temporal context for each vehicle and transaction. These include year, an integer representing the vehicle's model manufacturing year, and saledate, a timestamp detailing the exact day, date, and local time the auction occurred. Finally, the numeric measures quantify physical vehicle usage and valuation metrics. Continuous vehicle usage is tracked via odometer, which measures accumulated mileage,

alongside condition, a standardized pre-auction rating scored on a scale from 1.0 to 50.0. Financial benchmarks are measured through mmr, representing the wholesale Manheim Market Report valuation, and sellingprice, which records the finalized auction transaction price in U.S. dollars.

## **Required Transformations and Feature Engineering**

Regarding dates and time fields, the saledate attribute is currently recorded as an unstandardized text string that includes complex timestamp and timezone information. This field will require conversion into a standardized datetime format, enabling the extraction of granular temporal components such as transaction month, quarter, and day of week. Furthermore, this conversion will allow for the construction of a calculated vehicle age measure by subtracting the manufacturing year from the transaction year.

For categorical and text variables, columns such as body, make, and color display casing discrepancies and irregular formatting, such as overlapping entries for "Sedan" versus "sedan". These entries must undergo standard text normalization—converting values into uniform lower-case or title-case strings—to prevent identical vehicle segments and attributes from being split across redundant categories.

Finally, geographic and numeric fields present opportunities for feature preparation and derivation. The geographic variable state will be standardized to ensure clean regional groupings, while the financial measures sellingprice and mmr will be utilized to construct a derived price spread variable ( $\text{sellingprice} - \text{mmr}$ ). This derived measure will quantify the exact premium or discount realized on each vehicle relative to standard wholesale market benchmarks.

## **Observed Data-Quality Issues**

An initial audit of the data reveals several notable data-quality issues that will be addressed during the data cleaning phase. Foremost among these are missing values across several key variables, including transmission (~65,352 nulls, or roughly 11.7%), body (~13,195 nulls), condition (~11,820 nulls), make (~10,301 nulls), and model (~10,399 nulls). In addition, there are structural alignment issues where unescaped commas within seller or vehicle descriptions caused field values to shift into adjacent columns. These corrupted entries will be isolated and corrected or removed during preprocessing. Finally, numeric checks identified unusual values and extreme outliers, including transactions recorded at nominal values (such as \$1 sales) and zero-mileage odometer readings, both of which require auditing to prevent distortion in downstream analyses. Because all required variables originate from a single consolidated auction file (car\_prices.csv), cross-dataset merging was not required for this submission.